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# Cross-Model Interpretability via Sparse Feature Alignment between BERT and GPT-2

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## Abstract

Transformer-based classifiers achieve strong predictive performance but are difficult to interpret due to distributed internal representations. This work studies whether sparse autoencoder (SAE) features extracted from pretrained language model activations can be used to analyze classifier behavior in an interpretable way. We fine-tune a BERT-based classifier on the IMDB movie review dataset to obtain document-level predictions. We then extract residual stream activations from a pretrained GPT-2 model and decompose them into sparse latent features using a pretrained SAE (SAE Lens). To evaluate the relationship between SAE features and classification behavior, we apply three statistical measures: variance-based class separation, Welch’s t-test, and logistic regression coefficients. These signals are combined into a single ranking of feature importance. We find that many SAE features show statistically significant class separation, but rankings across different metrics are only weakly aligned, suggesting that feature importance is distributed across multiple perspectives. Qualitative analysis shows that highly aligned features tend to capture evaluative and semantic discourse, while weakly aligned features reflect more stylistic or templated patterns. Overall, the results suggest that SAE features capture structured semantic information that partially relates to classifier decisions but does not directly correspond to them.

## 1 Introduction

Large language models and transformer-based architectures have achieved strong performance across many natural language processing tasks, including text classification, question answering, and generation (Vaswani et al. [2017], Devlin et al. [2019], Brown et al. [2020]). However, their internal representations are difficult to interpret, as information is distributed across high-dimensional activation spaces (Belinkov [2022]).

This has motivated work in mechanistic interpretability, which aims to understand how neural networks represent information internally. A key challenge is feature superposition (Elhage et al. [2022]), where multiple concepts are encoded within overlapping activation dimensions, making individual neurons or features hard to interpret.

SAEs have recently been proposed as a way to address this issue (Cunningham et al. [2023], Bricken et al. [2023]). By mapping dense activations into a higher-dimensional sparse space, SAEs can separate overlapping representations into more interpretable features. Prior work shows that these features often correspond to semantic or syntactic concepts in transformer residual streams (Cunningham et al. [2023], Bricken et al. [2023], Gurnee and Tegmark [2023]).

Despite this progress, it remains unclear whether SAE features are aligned with downstream task behavior. In particular, while SAE features may be interpretable, it is not obvious whether they capture information relevant to classifier decision-making (Marks and Tegmark [2023]).

This work addresses this question using a cross-model analysis framework. We fine-tune a BERT-based classifier on the IMDB movie review dataset (Maas et al. [2011]) and obtain predictions on held-out data. We then extract residual stream activations from GPT-2 on the same inputs and decompose them using pretrained sparse autoencoders.

We analyze whether SAE features are associated with classifier predictions using variance-based separation, t-tests, and logistic regression. We also perform qualitative analysis of highly activating examples to understand the types of linguistic patterns captured by these features.

We ask the following research question:

**RQ:** *Do SAE features extracted from GPT-2 align with BERT classifier predictions, and what types of linguistic structure do aligned features represent?*

We hypothesize that strongly aligned features capture higher-level semantic and evaluative patterns, while weakly aligned features reflect more stylistic or affective signals. We expect this relationship to be graded rather than binary.

Our contributions are:

- A cross-model framework linking SAE-based interpretability with classifier behavior.
- Evidence that SAE features show weak but statistically significant alignment with predictions, varying across ranking methods.
- Qualitative analysis showing that strongly aligned features correspond to semantic discourse, while weakly aligned ones reflect stylistic or affective patterns.

Overall, the results suggest that SAE features encode structured semantic information that partially overlaps with classifier decision boundaries but does not directly determine them.

## 2 Related Work

### 2.1 Interpretability in Transformer Models

Transformer models achieve strong performance in NLP tasks but remain difficult to interpret due to distributed internal representations. The Linear Representation Hypothesis suggests that semantic concepts are encoded as directions in high-dimensional space rather than individual neurons (Park et al. [2023]).

A key challenge is feature superposition, where multiple concepts are encoded in overlapping directions. Elhage et al. [2022] formalized this phenomenon and showed that neural networks compress multiple features into shared representational spaces, limiting neuron-level interpretability.

Recent work in mechanistic interpretability focuses on identifying structured internal computations. For example, Ge et al. [2024] propose scalable methods for discovering circuits in transformer models. These approaches suggest that understanding model behavior requires feature-level rather than neuron-level analysis.

### 2.2 Sparse Autoencoders for Feature Discovery

SAEs have emerged as a method for decomposing transformer activations into more interpretable features. Cunningham et al. [2023] show that sparse dictionary learning applied to residual streams can recover semantically meaningful features and reduce polysemanticity. Kissane et al. [2024] extend this approach to additional transformer components, showing that sparse feature decompositions generalize across model internals. Large-scale evaluations further suggest that SAEs can scale to large language models while preserving reconstruction quality and sparsity Gao et al. [2024]. Related work also explores sparse approximations of transformer computation for circuit analysis Dunefsky et al. [2024].

Together, these results support SAEs as a practical tool for extracting interpretable latent features from transformer representations.

### 2.3 Evaluating Sparse Representations

A key open question is how to evaluate whether sparse features are useful beyond reconstruction and qualitative inspection. Most existing work focuses on intrinsic metrics such as sparsity, reconstruction error, or qualitative interpretability Shu et al. [2025]. However, these do not directly measure whether features are relevant to downstream tasks. O’Neill and Bui [2024] highlight the need for stronger functional validation of sparse features.

Universal sparse representations have also been explored in vision models Lan et al. [2024], but less work connects sparse features to supervised decision behavior in language models.

### 2.4 Cross-Model Interpretability

Most SAE-based studies analyze features within a single model. Less attention has been given to whether sparse features from one model align with decision behavior in another. This work addresses this gap by correlating SAE features extracted from GPT-2 with predictions from a fine-tuned BERT classifier. This enables evaluation of whether sparse features correspond to task-relevant structure rather than model-specific artifacts.

This approach is related to the Platonic Representation Hypothesis (Huh et al. [2024]), which suggests that different models converge toward shared representational structure. We test this idea by examining whether GPT-2 SAE features align with BERT decision boundaries.

This cross-model analysis allows us to move from descriptive interpretability toward behavioral validation of learned representations.

## 3 Methodology

### 3.1 Classification Model Fine-Tuning

We fine-tune a pretrained BERT model on the IMDB movie review dataset for binary sentiment classification (See table 1). The dataset contains natural language reviews labeled as positive or negative sentiment (Table 2). Input texts are tokenized using the BERT tokenizer and truncated to a maximum sequence length. The model is trained using cross-entropy loss with the AdamW optimizer (Figure 1).

| Parameter       | Value              |
|-----------------|--------------------|
| Model           | bert-base-uncased  |
| Learning rate   | $2 \times 10^{-5}$ |
| Batch size      | 32                 |
| Epochs          | 10                 |
| Sequence Length | 128                |

Table 1: Training Configuration of BERT

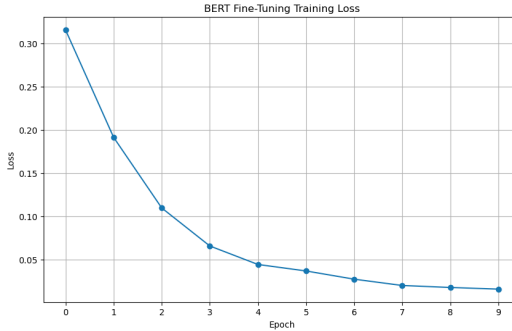


Figure 1: Training Loss During BERT Fine-Tuning

After training, the model generates predictions on the evaluation set. These predictions are used as labels for the SAE analysis, enabling comparison between sparse feature activations and classifier decision boundaries.

| Review Snippet                                                            | Label    |
|---------------------------------------------------------------------------|----------|
| "This film was emotionally compelling and beautifully acted..."           | Positive |
| "The plot was incoherent and the pacing made it nearly unwatchable..."    | Negative |
| "Strong performances partially redeem an otherwise predictable script..." | Positive |
| "Poor editing and weak dialogue make this difficult to recommend..."      | Negative |

Table 2: Representative examples from the IMDB sentiment classification dataset.

### 3.2 Sparse Autoencoder-Based Representation Decomposition

We use SAEs as a mechanism for decomposing dense transformer activations into interpretable latent features. A key challenge in mechanistic interpretability is that transformer hidden states are highly distributed: individual dimensions often participate in representing multiple overlapping concepts, a phenomenon commonly referred to as feature superposition.

Sparse autoencoders address this by projecting dense activations into a higher-dimensional latent space under sparsity constraints. Given a residual activation vector  $h \in \mathbb{R}^d$ , The encoder computes a sparse latent representation:

$$z = \text{ReLU}(W_e h + b_e)$$

where:

- $W_e$  is the encoder projection matrix
- $b_e$  is the encoder bias
- $z$  is the sparse latent code

The decoder reconstructs the original activation:

$$\hat{h} = W_d z + b_d$$

Training enforces that only a small subset of latent dimensions activate for any given input. This sparse decomposition provides two major interpretability advantages: Latent dimensions often exhibit selective activation patterns that can correspond to semantically or functionally coherent features in the input distribution, and Individual features can be analyzed independently through activation-based inspection. This project uses pretrained sparse autoencoders accessed through the SAE Lens framework (Bloom et al. [2024]). Specifically, we use the `gpt2-small-res-jb` checkpoint, a 24,576-feature sparse dictionary trained on residual stream activations from GPT-2 Small at layer 8 and made available through Neuronpedia (Lin-Perceley et al. [2023]). Transformer activations were extracted and manipulated using the TransformerLens interpretability library (Nanda and Bloom [2022]).

This configuration enables analysis of a high-capacity sparse feature dictionary while remaining computationally tractable. The objective is to determine whether these sparse latent dimensions capture interpretable concepts that correlate with classifier behavior. We focus on GPT-2 residual stream representations and BERT-based predictions to investigate whether different pretrained transformer models encode shared semantic structure that can be recovered through sparse feature decomposition.

### 3.3 Residual Stream Extraction

Each document is independently processed through GPT-2 to obtain intermediate residual stream activations. For a document of length  $T$ , the extracted hidden representation at layer  $l$  is

$$H^{(l)} \in \mathbb{R}^{T \times d}$$

where:

- $T$  denotes token sequence length

- $d$  denotes residual stream dimensionality

Residual activations are extracted specifically from layer 8. This layer was selected as an intermediate representational stage because this layer is commonly used in probing studies as it captures intermediate representations that reflect both lexical and higher-level contextual information. Unlike standard token-level analysis pipelines, document boundaries are explicitly preserved. This is necessary because downstream supervision is defined at the document level. Flattening activations across the corpus would destroy the correspondence between sparse features and classifier predictions.

### 3.4 Document-Level Sparse Representation Construction

Each token-level residual vector is encoded through the sparse autoencoder, producing token-level latent activations:

$$z_t \in \mathbb{R}^F$$

where  $F = 24,576$ .

To align these token-level activations with document-level labels, mean pooling is applied:

$$z_{\text{doc}} = \frac{1}{T} \sum_{t=1}^T z_t$$

This aggregation produces a single sparse feature vector for each document.

The resulting activation matrix is

$$Z \in \mathbb{R}^{N \times F}$$

where:

- $N$  is the number of documents
- $F$  is the number of sparse latent features

This matrix serves as the primary input for feature importance analysis.

### 3.5 Statistical Feature Ranking

The central analytical objective is to identify sparse latent features most strongly associated with classifier predictions. To ensure robustness, feature importance is evaluated using three complementary statistical criteria.

**Variance-Based Class Separation** For each feature, mean activation is computed separately for each predicted class. Importance is estimated as the variance across class-wise means. This measures the degree to which a feature discriminates between classes.

**Welch’s t-Test** Welch’s t-test evaluates whether activation differences between classes are statistically significant. Unlike the standard Student t-test, Welch’s formulation does not assume equal variance. This provides a more reliable estimate under heterogeneous activation distributions.

**Logistic Regression Coefficient Analysis** A logistic regression classifier is trained using sparse activations as predictors. The absolute magnitude of learned coefficients serves as a measure of predictive relevance. Features with larger coefficients contribute more strongly to class separation.

**Composite Importance Scoring** To combine complementary evidence, feature rankings are integrated into a unified score:

$$\text{Score} = \text{Variance} + |\beta| - \log(p)$$

where:

- Variance captures activation separability
- $|\beta|$  captures predictive contribution
- $p$  captures statistical significance

This composite metric reduces reliance on any single criterion and improves robustness of feature selection. Top-ranked features (Strong SAE) and bottom-ranked features (Weak SAE) are then selected for qualitative interpretation through inspection of highly activating documents.

### 3.6 Qualitative Feature Interpretation

To interpret selected SAE features, we perform manual inspection of highly activating documents. Features are ranked using the composite importance score defined in Section 3.5. Features with the highest scores are referred to as *strongly aligned*, while those with the lowest scores are referred to as *weakly aligned*.

For each selected feature, we retrieve the  $k = 5$  highest-activating documents from the evaluation set and inspect them for recurring semantic, stylistic, and structural patterns.

Interpretation follows an inductive coding procedure:

- Inspect highest-activation documents
- Identify recurring semantic or syntactic patterns
- Assign provisional labels
- Compare patterns across strongly and weakly aligned features
- Refine labels through iterative review

Features are grouped into the following interpretive categories:

- Topic-semantic features
- Sentiment and evaluative features
- Stylistic features
- Grammatical or syntactic features
- Mixed or ambiguous features

This analysis examines whether strongly aligned features tend to capture higher-level semantic structure, while weakly aligned features reflect lower-level stylistic or affective patterns.

## 4 Results

### 4.1 BERT and SAE Representations

We split the dataset into a training set (80%) and a test set (20%). The training set is used to fine-tune a BERT-based classifier, while the test set is used for evaluation. The model achieves a test accuracy of 0.89 (see Table 3 for full evaluation metrics).

| Metric                        | Value |
|-------------------------------|-------|
| Accuracy                      | 0.898 |
| Precision (Positive)          | 0.882 |
| Recall (Positive)             | 0.92  |
| F1-score (Positive)           | 0.90  |
| Specificity (Negative Recall) | 0.88  |
| Error Rate                    | 0.101 |

Table 3: Evaluation metrics for IMDB sentiment classification model

To analyze internal representations beyond BERT, we extract activations from a pretrained GPT-2 model. Specifically, we use the residual stream from layer 8 of GPT-2 small.

Let  $h \in \mathbb{R}^{T \times 768}$  denote the hidden states of a given input sequence, where 768 is the model hidden dimension and  $T = 128$  is the sequence length. We compute:

- GPT-2 model: GPT-2 small (12-layer transformer)
- Extracted layer: hidden state at layer 8
- Residual stream dimension: 768

These residual representations are then passed through a pretrained Sparse Autoencoder (SAE):

$$z = SAE(h)$$

where the SAE maps each 768-dimensional residual vector into a high-dimensional sparse feature space:

$$\mathbb{R}^{768} \rightarrow \mathbb{R}^{N_{features}}$$

For the GPT-2 SAE checkpoint used in this work, the feature dimension is:

$$N_{features} = [5000 \times 24576]$$

We aggregate token-level activations by mean pooling:

$$z_{doc} = \frac{1}{T} \sum_{t=1}^T z_t$$

This produces a document-level sparse representation where most feature activations are exactly zero or close to zero, reflecting the overcomplete and sparsity-inducing nature of the SAE representation (see Figure 2).

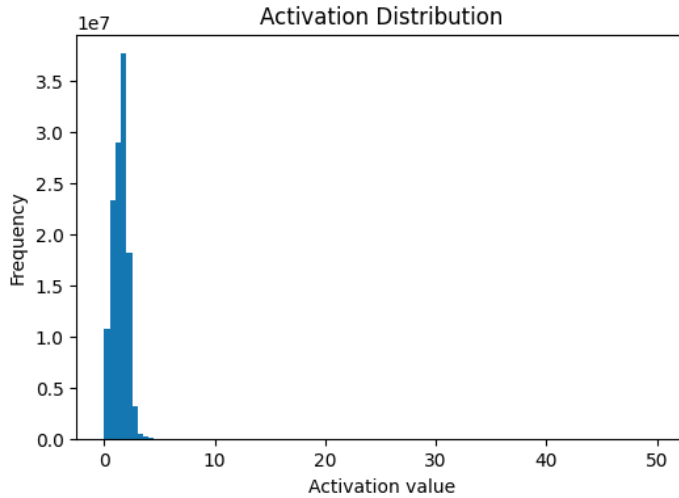


Figure 2: SAE Feature Activation Distribution

**Key Observation** Unlike BERT embeddings, which remain dense and low-dimensional, SAE representations decompose the GPT-2 residual stream into a large set of interpretable sparse features, enabling fine-grained analysis of semantic and stylistic structures in text.

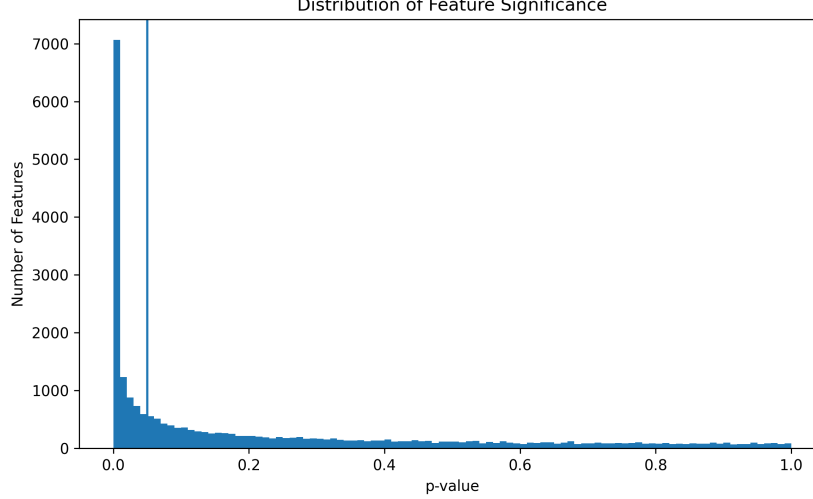


Figure 3: Distribution of feature-level statistical significance across SAE dimensions. The vertical threshold at  $p = 0.05$  indicates that a substantial proportion of latent features exhibit significant class separation.

## 4.2 SAE-BERT Feature Alignment Analysis

To quantify the relationship between SAE features and downstream classification behavior, we evaluate the extent to which document-level SAE activations align with predictions produced by the fine-tuned BERT classifier.

Let  $Z \in \mathbb{R}^{N \times F}$  denote document-level SAE activations, where  $N$  is the number of evaluation documents and  $F$  is the number of latent features. Let  $Y$  denote the predicted class labels produced by BERT.

Following the statistical feature-ranking framework introduced in Section 3.5, we evaluate feature relevance using variance-based separability, Welch’s t-test significance analysis, logistic regression coefficient magnitude, and a composite ranking score integrating all three criteria.

**Statistical Significance of SAE Features** Figure 3 shows the distribution of feature-level significance values across all latent dimensions. The distribution is strongly concentrated near lower significance values, indicating that a substantial subset of sparse latent dimensions exhibits reliable class-sensitive activation differences. At the same time, the presence of a long tail of non-significant features suggests that task-relevant information is unevenly distributed across the SAE feature space rather than uniformly encoded.

**Cross-Metric Agreement** Agreement across ranking criteria is summarized in Figure 4. The asymmetry in overlap values suggests that the composite ranking score more closely tracks predictive structure than purely variance-based ranking. In particular, the relatively stronger overlap between logistic-regression-based and composite rankings indicates that predictive contribution plays a dominant role in the integrated scoring framework.

**Relationship Between Statistical and Predictive Importance** The relationship between variance-based separability and logistic regression importance is visualized in Figure 5. Although the overall trend is positive, the substantial dispersion of points indicates that statistical separability and predictive contribution are only partially aligned. Many features with strong class-wise separation exhibit limited predictive influence, while some highly predictive features display comparatively modest variance separation.

### Quantitative Summary

- Number of significant features ( $p < 0.05$ ): **10,496**



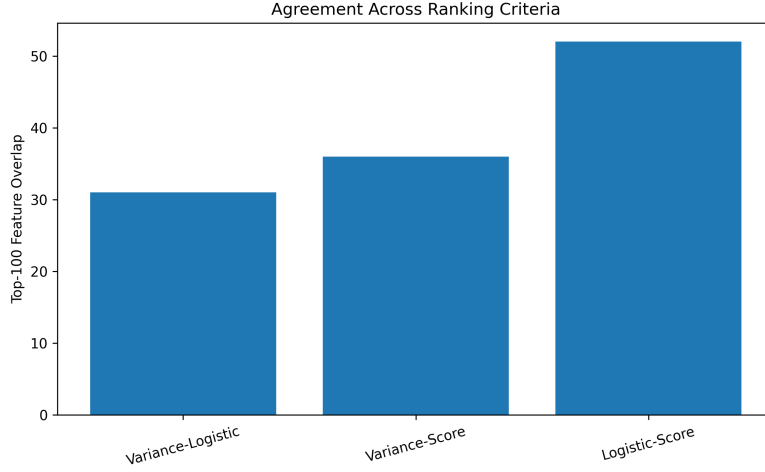


Figure 4: Overlap among top-100 SAE features selected by different ranking methods. Limited overlap suggests that each metric captures distinct aspects of feature relevance.

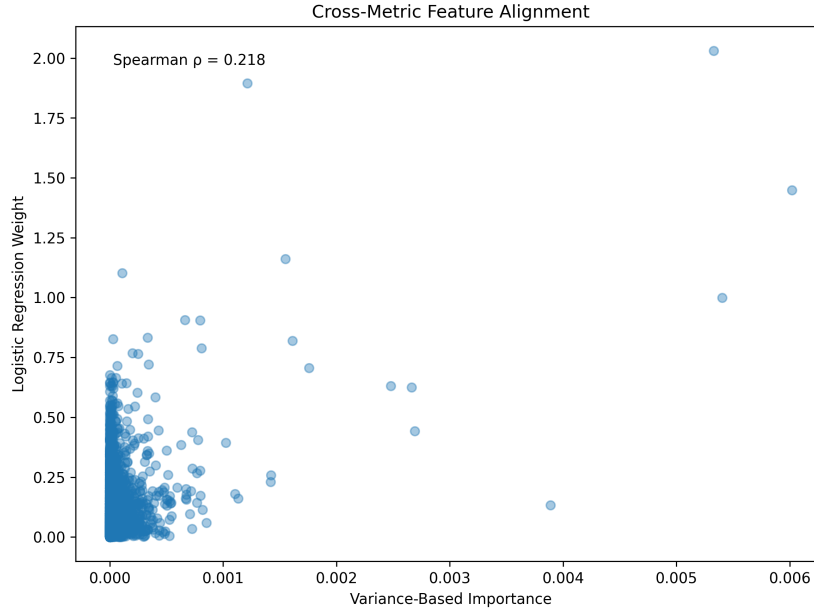


Figure 5: Relationship between variance-based separability and logistic regression importance across SAE features. The weak but positive correlation indicates partial alignment between statistical separability and predictive contribution.

- Top- $k$  overlap between ranking methods ( $k = 100$ ):  
**Variance-Logistic = 31, Variance-Score = 36, Logistic-Score = 52**
- Correlation between variance and logistic weights (Spearman  $\rho$ ):  
**0.235** ( $p < 0.001$ )

**Key Observation** Using an evaluation set of  $N = 5000$ , we observe moderate but incomplete consistency across feature-ranking criteria. Approximately 42.7% of all sparse latent dimensions (10,496 out of 24,576) exhibit statistically significant class separation, indicating substantial alignment between SAE activations and downstream classifier behavior.

However, this alignment is distributed rather than concentrated. The overlap between ranking methods is non-trivial but incomplete, with the strongest agreement emerging between logistic-regression-based and composite rankings. Variance-based rankings show weaker overlap with predictive rankings, suggesting that statistical separability and predictive contribution capture related but distinct aspects of feature importance.

Additionally, variance-based importance and logistic regression weights exhibit a weak but statistically significant positive correlation ( $\rho = 0.235, p \ll 0.001$ ), reinforcing the view that these metrics capture partially overlapping but non-identical dimensions of feature relevance.

Taken together, these findings indicate that SAE feature importance is inherently multi-dimensional rather than reducible to a single dominant ranking criterion.

**Interpretation Shift** Initial experiments conducted on smaller evaluation subsets suggested substantially stronger apparent agreement between ranking methods. However, scaling evaluation to larger sample sizes revealed that feature alignment is more distributed and heterogeneous than initially expected.

This shift highlights the sensitivity of sparse feature analysis to dataset scale and emphasizes the importance of evaluating interpretability claims on sufficiently large corpora.

| Feature ID | Label                                                       | Category                                       | Evidence                                                                       |
|------------|-------------------------------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------|
| 18435      | Long-form evaluative media review                           | Topic-semantic + Sentiment                     | Movie reviews with emotional judgment, comparison, reflective critique         |
| 20528      | Film aesthetics and cinematic analysis                      | Topic-semantic                                 | Cinematic style, technicolor, voyeurism, directorial interpretation            |
| 18337      | Critical comparative media analysis                         | Topic-semantic + Evaluative                    | Film comparisons, narrative critique, ideological framing                      |
| 6267       | Horror/cult memory-based evaluative critique                | Topic-semantic + Affective memory + Evaluative | Childhood memory, horror imagery, genre comparison, writing quality critique   |
| 18777      | Light evaluative entertainment discourse                    | Topic-semantic + Sentiment                     | Positive film enjoyment, comedy evaluation, comparative film judgment          |
| 15158      | Narrative-driven film appreciation with structural critique | Topic-semantic + Evaluative                    | Plot progression, cinematography appreciation, formulaic storytelling analysis |
| 19829      | Sensory/emotional evaluative media discourse                | Topic-semantic + Affective + Evaluative        | Strong emotional reactions, moral/sexual framing, film judgment                |
| 13934      | Low-budget film criticism and meta-commentary               | Evaluative + Meta-discourse                    | Production critique, technical failure, sarcastic evaluation                   |
| 8631       | Negative evaluative entertainment rejection                 | Sentiment + Evaluative                         | Strong rejection, humor critique, dismissive audience judgment                 |
| 3071       | Mixed evaluative narrative review                           | Topic-semantic + Evaluative                    | Balanced critique, plot confusion, partial appreciation                        |

Table 4: Codebook of top SAE features and qualitative interpretation

### 4.3 Strong SAE Features: Semantic and Evaluative Structure

Strong and weak features are selected based on the composite importance score ranking (top-10 and bottom-10 features respectively). Based on the score the top 10 most strongly activated SAE features: 18435, 20528, 18337, 6267, 18777, 15158, 19829, 13934, 8631, and 3071. Table 4 summarizes their labels, semantic categories, and qualitative evidence derived from highly activating documents.

A clear pattern emerges across these features: SAE latents are not capturing isolated lexical or syntactic properties, but instead encode structured discourse types commonly found in film review communities. These include evaluative reasoning, aesthetic interpretation, comparative critique, and affective narrative reflection.

Feature 18435 strongly activates on long-form evaluative discourse in which users express subjective judgments about films or media content. This feature captures both positive and negative sentiment, combined with comparative reasoning and reflective interpretation. Rather than representing simple polarity, it reflects structured opinion formation grounded in narrative and character evaluation. For example:

*“As kids movie it is great. For the family it just sucks. I was truly hoping for something like the Goonies which is a great film for all ages...”*

This suggests that the feature encodes a higher-level semantic concept associated with media evaluation discourse, rather than purely sentiment or lexical cues.

Feature 20528 encodes discourse focused on cinematic aesthetics and film theory, including references to visual style, directorial influence, and interpretive commentary on formal techniques. Unlike general sentiment features, this latent captures domain-specific analytical language used in film criticism. For instance:

*“Brilliant use of overstated technicolor illustrates the optimistic extremes of present day Christmas ceremonies... the voyeuristic element... is unique...”*

This indicates sensitivity to stylistic and formal properties of film rather than narrative content alone.

Feature 18337 captures structured comparative critique in which users evaluate films based on narrative coherence, ideological framing, and relative quality across different works. This feature reflects higher-order evaluative reasoning that integrates comparison and argumentation rather than isolated opinion. For example, users critique narrative plausibility and ideological intent, often contrasting films such as *Die Hard* or other genre benchmarks.

Feature 6267 represents a distinct discourse mode characterized by affective memory and genre-anchored interpretation. Users frequently situate their evaluation within childhood viewing experiences, particularly in horror and cult cinema contexts. This feature combines autobiographical memory with evaluative judgment and genre comparison. Unlike purely sentiment-based latents, it reflects long-term cognitive associations between early media exposure and present-day critical interpretation. It also frequently encodes normative claims about artistic quality, particularly emphasizing writing as a core determinant of cinematic value.

Feature 18777 activates on light evaluative entertainment discourse, typically associated with comedy and mainstream film appreciation. The language is characterized by accessible judgment, audience-oriented recommendations, and relatively low analytical depth compared to other features. However, it still reflects comparative evaluation across films and performers, indicating structured but less analytically dense critique.

Feature 15158 captures narrative-driven film appreciation combined with structural critique. Activations include discussion of plot progression, cinematography, pacing, and production design. While evaluative in nature, this feature emphasizes storytelling mechanics and cinematic execution, often acknowledging formulaic genre structures while still expressing overall appreciation.

Feature 19829 reflects strongly affective and morally charged evaluative discourse. It is characterized by heightened emotional language and explicit engagement with sensitive or provocative thematic content. This suggests that the feature encodes emotionally intensified interpretation of media, often tied to moral or social framing.

Feature 13934 captures meta-level criticism of low-budget or poorly executed films. The discourse often includes technical evaluation of production quality, special effects, and writing, frequently accompanied by sarcasm or dismissive commentary. This indicates sensitivity to production constraints and perceived artistic failure.

Feature 8631 represents strongly negative evaluative discourse characterized by outright rejection of media content. The language is typically dismissive, humorous, or sarcastic, often focusing on lack of entertainment value or perceived incompetence in filmmaking.

Feature 3071 encodes mixed evaluative narratives in which users simultaneously express confusion, partial appreciation, and critique. These reviews often reflect ambivalent reception, combining narrative critique with acknowledgment of some redeeming qualities.

**Summary of Feature Space Structure** Across all top activating features, we observe a consistent organization of SAE latents into interpretable discourse regimes. Rather than encoding surface-level lexical patterns, the SAE appears to capture higher-order semantic structures in film-related discourse. These include (i) evaluative sentiment reasoning, (ii) aesthetic and formal analysis, (iii) comparative narrative critique, and (iv) affective and memory-driven interpretation.

This suggests that sparse autoencoder features operating on transformer residual streams encode meaningful and semantically coherent units of discourse, particularly in structured domains such as film criticism.

| Feature ID | Label                                                    | Category                            | Evidence                                                                                        |
|------------|----------------------------------------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------|
| 17730      | Hyperbolic negative film appraisal                       | Topic-semantic + Sentiment          | extreme negative judgments, global film condemnation, "waste of money" framing                  |
| 6787       | Hyperbolic negative film appraisal (template repetition) | Topic-semantic + Sentiment          | repeated extreme criticism, identical structural negativity, video game adaptation critique     |
| 12205      | Nostalgic media description with low engagement          | Topic-semantic + Sentiment          | cartoon chase narrative summary, cultural sensitivity reference, low evaluative depth (4/10)    |
| 23053      | Hyperbolic negative film appraisal (redundant instance)  | Topic-semantic + Sentiment          | duplicate extreme negative review, global rejection of film quality, editing/story criticism    |
| 13390      | Comparative narrative and suspense evaluation            | Topic-semantic + Evaluative         | detective comparison, suspense scene analysis, war film interpretive framing                    |
| 6983       | Campy horror enjoyment and ironic appreciation           | Sentiment + Entertainment           | zombie gore descriptions, "bad but fun" framing, party viewing experience                       |
| 10112      | Biographical performance appreciation                    | Topic-semantic + Positive sentiment | actor performance praise, emotional biopic interpretation, character empathy and awards mention |
| 23975      | Enthusiastic fandom and serial TV appreciation           | Sentiment + Fandom discourse        | strong endorsement, DVD rewatch behavior, cancellation regret and fan loyalty                   |
| 14634      | Mixed evaluative action film critique                    | Topic-semantic + Evaluative         | technical assessment of plot and acting, balanced praise and criticism, action genre framing    |
| 18649      | Emotion-driven negative character judgment               | Sentiment + Affective reaction      | strong dislike of characters, disgust-based reaction, memory-linked emotional recall            |

Table 5: Codebook of weak SAE features and qualitative interpretation

#### 4.4 Weak SAE Features: Semantic and Evaluative Structure

We define weak features as the lowest-ranked features according to the composite importance score. Our identified weak SAE features: 17730, 6787, 12205, 23053, 13390, 6983, 10112, 23975, 14634, 18649. Table 5 summarizes their labels, semantic categories, and qualitative evidence derived from highly activating documents.

Feature 17730 captures highly hyperbolic negative film appraisal, characterized by extreme global judgments of cinematic failure. The discourse collapses multiple dimensions of evaluation (story, editing, directing) into a single overwhelming negative verdict, often expressed through exaggerated rating scales and dismissive language. The feature reflects structured total rejection discourse rather than nuanced critique. For example, *"I'd give it a -4... Everything from the story to directing is awful... don't watch it."*

Feature 6787 reflects a templated form of extreme negative evaluation, structurally similar to Feature 17730 but with more repetitive rhetorical framing. The discourse follows a stable pattern of condemnation, often referencing prior knowledge gaps (e.g., source material awareness) to justify negativity. This suggests a learned or formulaic style of strongly negative review writing.

Feature 12205 captures low-engagement nostalgic media description, where users summarize content (often animated or older media) with minimal analytical depth. The discourse includes cultural sensitivity awareness and simple evaluative scoring, but lacks structured critique. The feature reflects descriptive recall rather than analytical judgment.

Feature 23053 is a redundant instance of hyperbolic negative appraisal, closely mirroring Feature 17730 in structure and rhetorical intensity. It reinforces the presence of a stable negative review template in the dataset, suggesting repeated encoding of extreme dissatisfaction across multiple embeddings.

Feature 13390 encodes comparative narrative and suspense analysis in classic cinema contexts. The discourse involves cross-film comparison (e.g., detective figures, war cinema), alongside interpretive evaluation of narrative tension and thematic depth. Unlike purely emotional reviews, this feature reflects moderate analytical reasoning about storytelling and genre conventions.

Feature 6983 captures campy horror appreciation and ironic enjoyment of low-budget media. Despite recognizing poor production quality, users express enjoyment through humor, group viewing (“party tape” contexts), and exaggerated gore descriptions. This reflects a dual-layer sentiment structure: criticism coexisting with entertainment value.

Feature 10112 represents biographical film appreciation grounded in performance evaluation and emotional engagement. The discourse highlights actor transformation, narrative inspiration, and character empathy, often praising acting intensity and awards recognition. The feature encodes strong positive sentiment combined with interpretive admiration.

Feature 23975 captures fan-driven serial television enthusiasm, characterized by emotional attachment, rewatch behavior, and frustration over cancellation. The discourse reflects community-based valuation, where narrative quality is reinforced by fandom identity and continuity expectations.

Feature 14634 reflects mixed evaluative discourse in action cinema, combining technical critique (plot, writing, acting) with selective praise (setting, performance). The feature encodes balanced judgment rather than polarity extremes, suggesting structured but moderate analytic engagement.

Feature 18649 encodes emotion-driven negative character evaluation, where users express strong affective reactions such as disgust, dislike, or discomfort. The discourse is less about cinematic structure and more about moral or emotional rejection of characters and experiences, often reinforced by memory recall.

**Summary of Weak Feature Space Structure** Across the weak-feature set, we observe a consistent shift away from structured analytical reasoning toward extreme affective expression, templated review patterns, and emotionally driven judgment. Unlike the strong features, which encode domain-specific critique and comparative reasoning, weak features are characterized by: (i) Hyperbolic global evaluations (e.g., total film failure narratives) (ii) Repetitive or templated review structures (iii) Nostalgic or low-engagement descriptive summaries (iv) Strong emotional reactions (disgust, enjoyment, frustration) (v) Reduced analytical depth and limited conceptual abstraction.

Interestingly, some weak features still contain evaluative language; however, this evaluation is typically non-structured, affect-dominant, or formulaic, indicating that the model captures sentiment intensity more than interpretive reasoning in these dimensions.

Overall, the distinction between strong and weak features suggests that SAE importance rankings align with differences in discourse structure, where higher-ranked features tend to reflect more structured evaluative reasoning compared to lower-ranked features which more often exhibit templated or affect-driven expression.

#### 4.5 Ambiguous and Mixed Features

Some SAE features exhibit mixed characteristics, combining both semantic and structural signals. These features do not clearly align with either high-level evaluative discourse or low-level affective templates. This suggests that SAE latents may represent partially disentangled factors of discourse rather than strictly separable categories.

## 5 Discussion

### 5.1 Multi-View Structure of SAE Feature Importance

Our results suggest that sparse autoencoder features extracted from GPT-2 residual streams exhibit a multi-view structure with respect to downstream classification behavior. While a large proportion of features (10,496 out of 24,576) show statistically significant class separation, the agreement between different importance measures is limited.

In particular, variance-based ranking, logistic regression coefficients, and combined statistical scores identify partially overlapping but distinct sets of top features, with limited overlap between variance- and logistic-based rankings. This indicates that different analytical criteria capture distinct aspects of the SAE representation space rather than a unified notion of “importance.”

### 5.2 Weak but Significant Alignment with Classifier Behavior

Despite this divergence, we observe a statistically significant but weak-to-moderate positive correlation between variance-based feature importance and logistic regression weights ( $\rho = 0.218$ ,  $p \ll 0.001$ ). This suggests that while there is a global tendency for features that separate classes to also contribute to linear prediction, this relationship is not strong at the level of individual features.

This finding is consistent with the hypothesis that SAE representations are highly overcomplete and distributed, such that task-relevant information is spread across many partially redundant features.

### 5.3 Evaluation of the Feature-Semantics Hypothesis

A central hypothesis of this work was that sparse latent features most strongly aligned with classifier outputs would encode higher-level semantic concepts, whereas weakly aligned features would capture lower-level linguistic or stylistic structure.

Our qualitative analysis provides partial support for this hypothesis. Highly ranked features consistently activated on semantically rich evaluative discourse, including comparative film critique, aesthetic analysis, narrative interpretation, and structured opinion formation. These features reflect abstract discourse-level reasoning rather than isolated lexical triggers.

In contrast, weakly ranked features more frequently captured repetitive rhetorical templates, emotionally exaggerated reactions, nostalgic descriptive summaries, and formulaic evaluative expressions. While these features were not purely grammatical or syntactic in nature, they generally exhibited reduced conceptual abstraction and weaker semantic organization.

This suggests that classifier-aligned sparse features tend to correspond to higher-order semantic organization, while weakly aligned features more often reflect shallow stylistic or affective regularities. However, the distinction is not absolute, indicating that semantic interpretability exists along a continuum rather than as a binary separation.

### 5.4 Implications for Interpretability

These results challenge the assumption that sparse autoencoder features correspond to a single, stable ranking of semantic importance. Instead, SAE latents appear to encode a rich but non-uniform feature space where interpretability depends strongly on the chosen analytical lens.

In this setting:

- Variance captures discriminative structure across classes,
- Logistic regression captures predictive linear directions,
- t-tests capture distributional shifts,
- Combined scoring integrates these but does not reconcile them.

Therefore, interpretability of SAE features should be understood as multi-dimensional rather than scalar.

## 5.5 Relation to BERT Classifier Signals

Importantly, the observed partial alignment between SAE features and BERT predictions suggests that GPT-2 residual stream structure contains signals relevant to downstream classification tasks. However, this alignment is indirect and distributed, rather than being concentrated in a small subset of highly predictive features.

This supports the view that SAE decomposition reveals latent semantic structure in transformer representations, but does not yield a direct one-to-one mapping with classifier decision boundaries.

## 5.6 Limitations and Future Directions

While the results demonstrate meaningful alignment between sparse autoencoder features and downstream classifier behavior, several limitations should be noted.

First, the current analysis primarily relies on linear and univariate statistical measures, including variance-based separation, Welch’s t-test, and logistic regression coefficients. These approaches may not fully capture nonlinear or higher-order interactions between SAE features and classifier predictions. As a result, some complex representational dependencies in the SAE feature space may remain unexplored.

Second, this study uses GPT-2 small as the underlying generative model for residual stream extraction. Although GPT-2 provides a well-studied and computationally tractable setting, it is significantly less capable than modern large language models such as Gemma or other more recent transformer architectures. Consequently, the representational richness of the extracted residual stream may be limited compared to more advanced models.

Third, the analysis is constrained by a relatively small set of qualitative inspections (top-k activating documents per feature). While this provides interpretability insights, it does not scale to fully characterize the entire SAE feature space.

Future work could address these limitations in several ways. First, nonlinear analysis methods, such as mutual information estimators or kernel-based methods, could be used to better capture complex dependencies between SAE features and classifier behavior. Second, extending the analysis to larger and more capable language models would allow evaluation of whether the observed structural patterns generalize across model scales. Third, qualitative analysis could be partially automated or augmented using clustering methods or embedding-based grouping of high-activation documents, enabling more systematic exploration of feature semantics at scale.

## 6 Conclusion

This work investigates whether sparse autoencoder (SAE) features extracted from GPT-2 residual streams align with downstream classifier behavior. We propose a cross-model analysis framework that relates SAE latent activations to BERT-based predictions using multiple complementary statistical criteria, including variance-based separation, Welch’s t-test, and logistic regression coefficients.

Our results show that SAE feature importance is not consistent across ranking methods, revealing a multi-view structure in the learned representation space. While a large fraction of features exhibit statistically significant class separation, agreement between ranking criteria remains limited, and feature-level correlations are weak. This indicates that no single importance metric fully captures the structure of the SAE feature space.

Despite this inconsistency, qualitative analysis reveals a stable semantic pattern: strongly aligned features tend to capture structured evaluative and discourse-level reasoning, whereas weakly aligned features more often reflect templated, affect-driven, or less compositional language patterns. This suggests that interpretability emerges not from individual ranking signals alone, but from the interaction between statistical selection criteria and qualitative inspection.

Overall, our findings support the view that sparse autoencoder representations encode distributed and multi-faceted semantic structure that partially reflects classifier decision boundaries, but does not reduce to a single predictive decomposition. This highlights both the promise and the limitations of SAE-based interpretability for understanding transformer representations.

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